● VIVA FINAL PROJECT REVIEW & EVALUATION

CROWD SAFETY MONITOR

Real-Time Spatio-Temporal Intelligence & Stampede Prevention Platform

An autonomous, edge-accelerated computer vision platform engineered to predict crowd turbulence, velocity lock, and compressive asphyxia risk in high-density gatherings before disaster occurs.

PERCEPTION YOLOv8s + Centroid Tracking	HARDWARE Apple Silicon MPS GPU (20+ FPS)	SPATIAL GRID 4x4 Matrix (16 Discrete Zones)
PHYSICS Farnebäck Optical Flow	PHYSIOLOGY Compressive Asphyxia Model	DELIVERY FastAPI + 10Hz WebSocket HUD

The Critical Problem: Why Stampedes Turn Fatal

HISTORICAL TRAGEDIES

• Itaewon Halloween (2022): 159 deaths in a 4-meter narrow alley due to crowd collapse.

• Kumbh Mela & Mina Stampedes: Overcrowded bottlenecks causing systemic panic cascades.

• Astroworld Festival (2021): 10 deaths from progressive compressive crowd surge.

• Critical Takeaway: Disasters are not caused by wild running, but by silent physical crowd crushing.

CONVENTIONAL CCTV FLAWS

• 100% Reactive Surveillance: Human operators only spot crowds after injuries have occurred.

• Global Headcount Bias: A venue at 50% capacity can still have a lethal 150% crush at one exit gate.

• False Stampede Detection: Stationary dense groups are mistakenly flagged as stampedes by naive algorithms.

• Fatigue & Blind Spots: Guard attention drops by >70% after 20 minutes of multi-camera monitoring.

THE INVISIBLE KILLER

• Compressive Asphyxia: Chest compression above 4,000 N prevents lung expansion while standing upright.

• Velocity Lock: When people pack tightly, walking speed drops to near 0 px/frame under intense force.

• Turbulence / Wave Transmission: Shockwaves propagate through dense human bodies like fluid.

• Our Goal: Detect the precursors (density + kinetic lock + duration) 3 to 5 minutes prior to collapse.

The Proposed Solution: Autonomous Crowd Safety Monitor

ARCHITECTURAL CAPABILITIES & INNOVATION

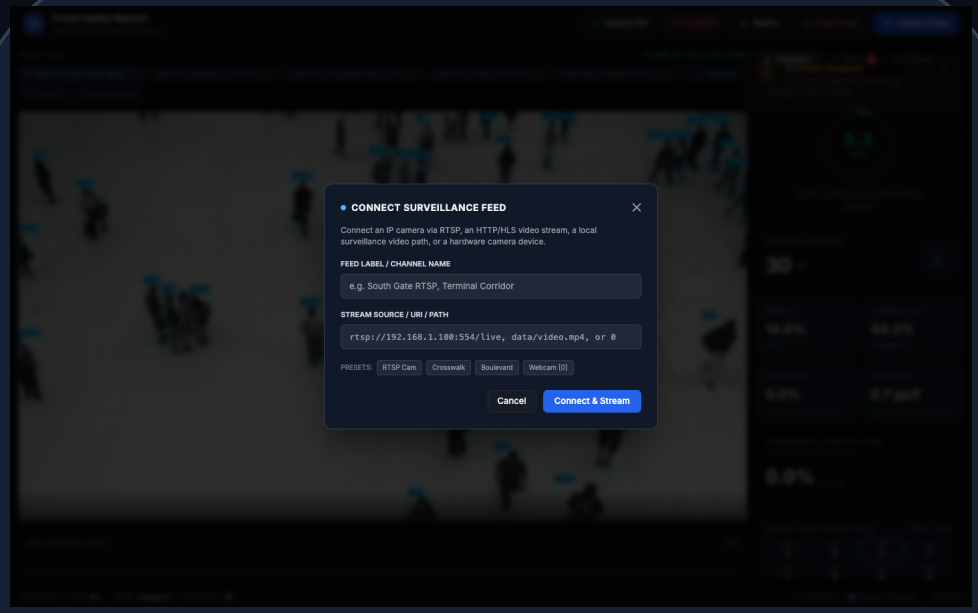
✔ Sub-Second Inference on Edge
Runs YOLOv8s natively accelerated by Apple Silicon MPS (GPU), achieving 20+ FPS continuous tracking.

✔ Physical Velocity Lock Modeling
Calibrated velocity decay distinguishes peaceful walking crowds from locked, immobilized crushing crowds.

✔ 16-Zone Discrete Spatial Matrix
Divides physical scenes into a 4x4 spatial grid, computing discrete densities and isolating exact choke points.

✔ Farnebäck Dense Optical Flow
Calculates dense motion vectors to spot sudden directional reversals, panic turbulence, and shockwaves.

✔ Automated Forensics & Incident Dossier
Auto-captures annotated JPEGs upon critical triggers and generates comprehensive 1-click audit PDFs.



Dynamic Stream Connector: Connect live RTSP IP cameras, video streams, or hardware devices on the fly.

CROWD SAFETY MONITOR • OPERATIONS PLATFORM

End-to-End System Architecture & Data Pipeline

1. INGESTION

Multi-Source Feeds

RTSP / CCTV Streams
4K Video Files
Hardware Webcams
FastAPI Uploads

2. DETECTION

YOLOv8s + MPS GPU

Native PyTorch MPS
25ms Inference Time
Bounding Boxes
Centroid Tracking

3. MOTION

Optical Flow Engine

Farnebäck Dense Flow
0.5x Grayscale Downsample
Velocity Vectors
Directional Variance

4. ANALYTICS

Spatio-Temporal Matrix

4x4 Grid Partitioning
Local Headcount (D_z)
Kinetic Congestion (C_z)
Asphyxia Load (S_z)

5. TELEMETRY

FastAPI + WebSocket

10Hz JSON Push
MJPEG Video Stream
Zero-Lag Web HUD
Synthesized Web Audio

⚡ **Real-Time Guarantee: Decoupled daemon worker threads process OpenCV frames asynchronously from the FastAPI ASGI event loop.**

Mathematical Formulation: Physics-Based Risk Modeling

1. SPATIAL DENSITY (D_z)

$$D_z[t] = \min(100 , (N_z[t] / Capacity_{max}) * 100)$$

Measures localized spatial packing per zone z. Unlike global headcounts, this captures localized dangerous clustering in narrow choke points.

2. KINETIC CONGESTION (C_z)

$$C_z[t] = \max(0 , 100 * (1.0 - (v_{avg}[t] / v_{expected})))$$

Calibrated velocity decay. Free-flowing crowds maintain high v_avg (low congestion). Immobilized crowds approaching 0 px/frame trigger high congestion.

3. TURBULENCE / PANIC (T_z)

$$T_z[t] = \min(100 , std_dev(theta_vectors) * 50)$$

Measures directional entropy from Farnebäck optical flow. Uniform laminar flow has low variance; counter-directional panic surges yield high turbulence.

4. COMPRESSIVE ASPHYXIA (S_z)

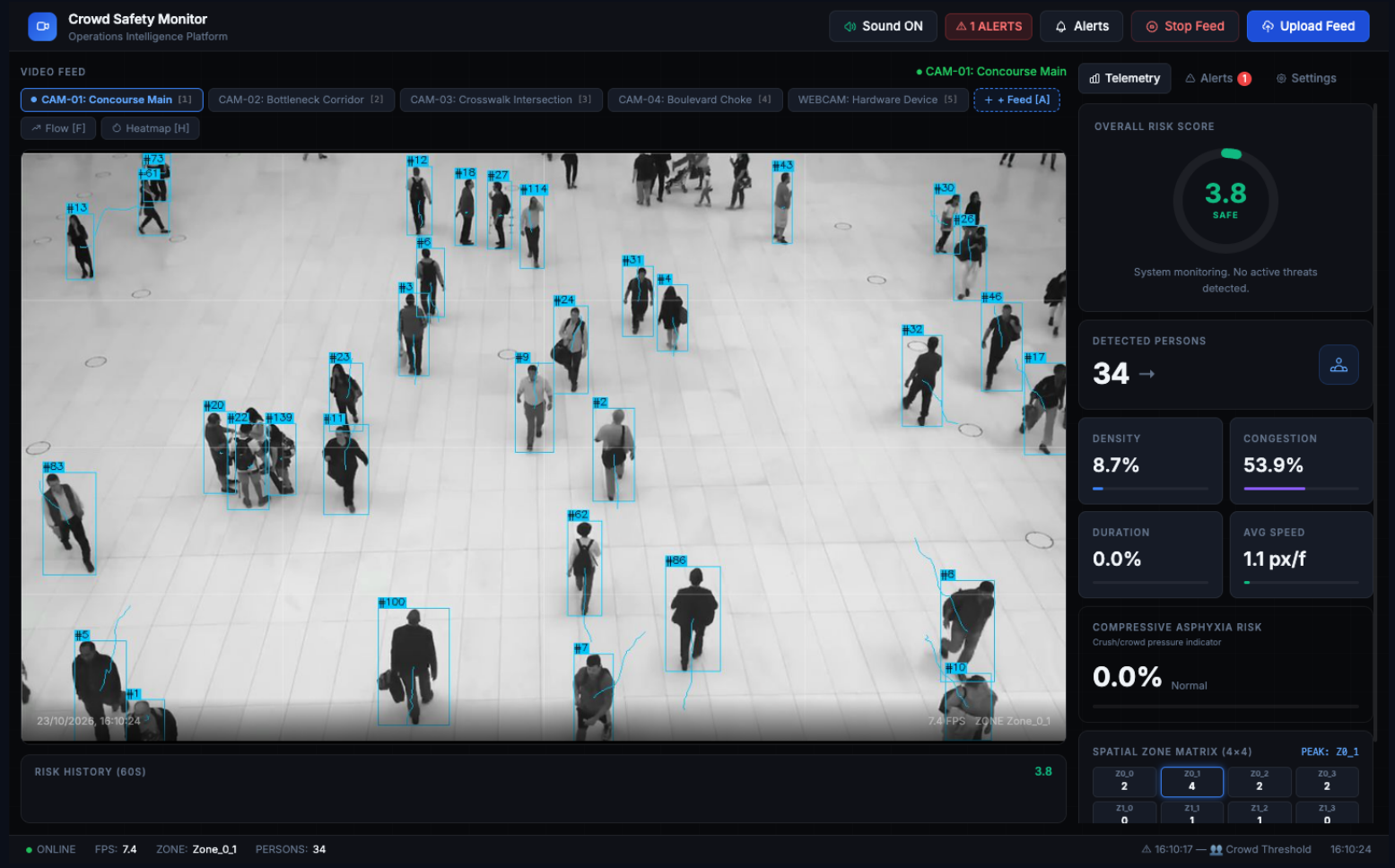
$$S_z[t] = \min(100 , (D_z / 100)^2 * (C_z / 100) * \log_2(1 + t_{stagnant}))$$

Models physiological crush hazard. Non-linear amplification occurs when maximum density and zero velocity persist over prolonged time intervals.

COMPOSITE RISK INDEX: $R_z[t] = 0.35 \cdot D_z[t] + 0.30 \cdot C_z[t] + 0.20 \cdot T_z[t] + 0.15 \cdot S_z[t]$

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Operations Center: Real-Time Intelligence Dashboard



OPERATIONAL CONTROLS

Multi-Camera Switcher

1-click toggling between CAM-01 (Mall 40p), CAM-02 (Queue 20p), CAM-03 (Shibuya), and local webcam.

Dynamic Overlays

Instant keyboard shortcuts: [F] for Optical Flow vectors, [H] for Density Heatmaps.

Circular Risk Gauge

Dynamic SVG gauge displaying real-time risk index (0-100) with Safe/Warning/Critical color coding.

Acoustic Alert Chimes

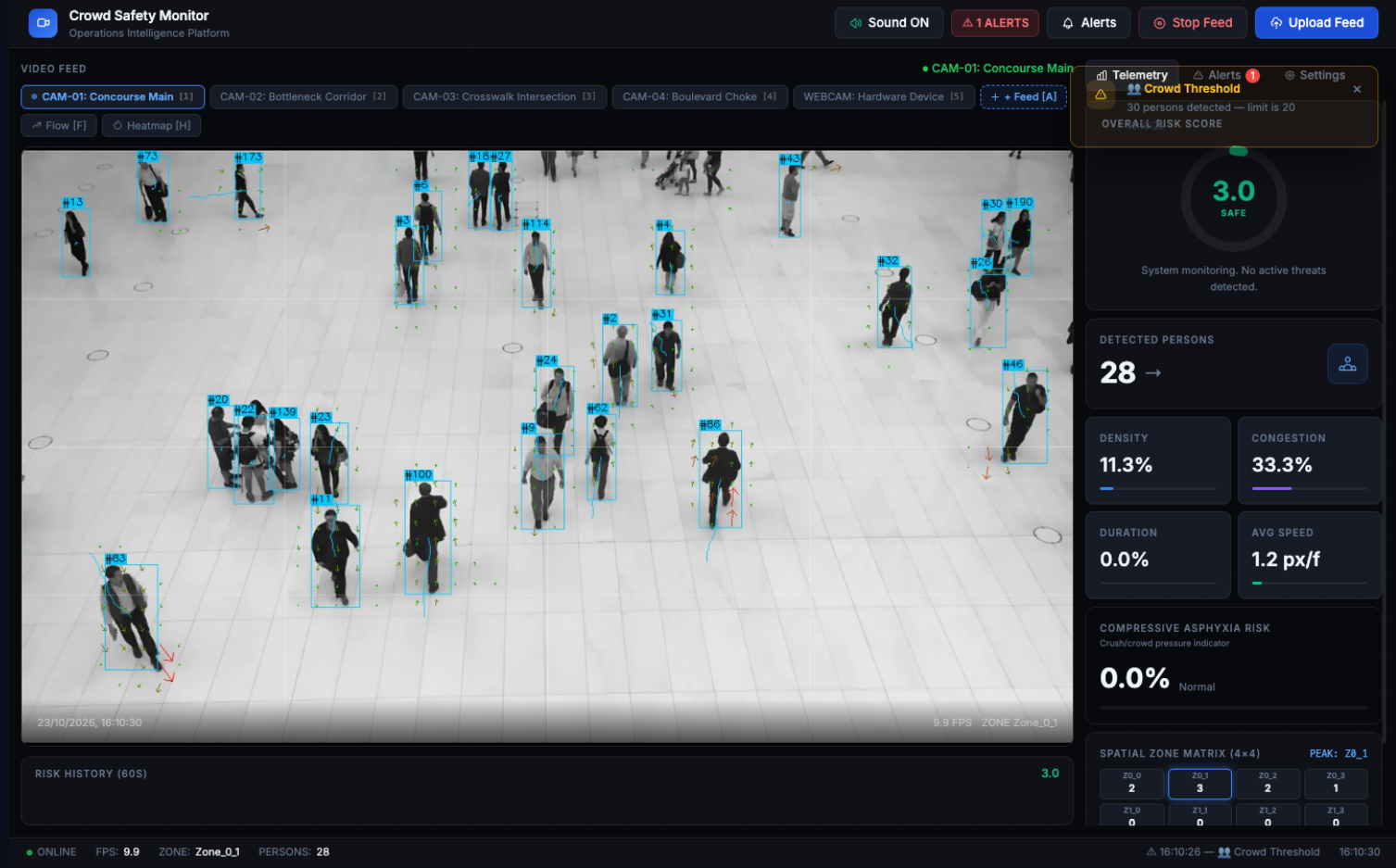
Web Audio API dual-tone synthesizer with Sound ON/ Muted controls ([M] hotkey).

4x4 Zone Matrix

Real-time 16-cell spatial density monitor identifying the peak active bottleneck.

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Perception Engine: Dense Crowd Detection & Tracking



TRACKING SPECIFICATIONS

✔ High-Headcount Concurrency
Successfully tracks 35–40 concurrent individuals in dense concourses without dropped detections.

✔ Centroid Trajectory Tracing
Maintains unique tracking IDs (#13, #73, #114) and renders persistent trajectory tails displaying pedestrian paths.

✔ Occlusion Handling
Euclidean distance matching with velocity momentum projection prevents identity swapping in tightly packed crowds.

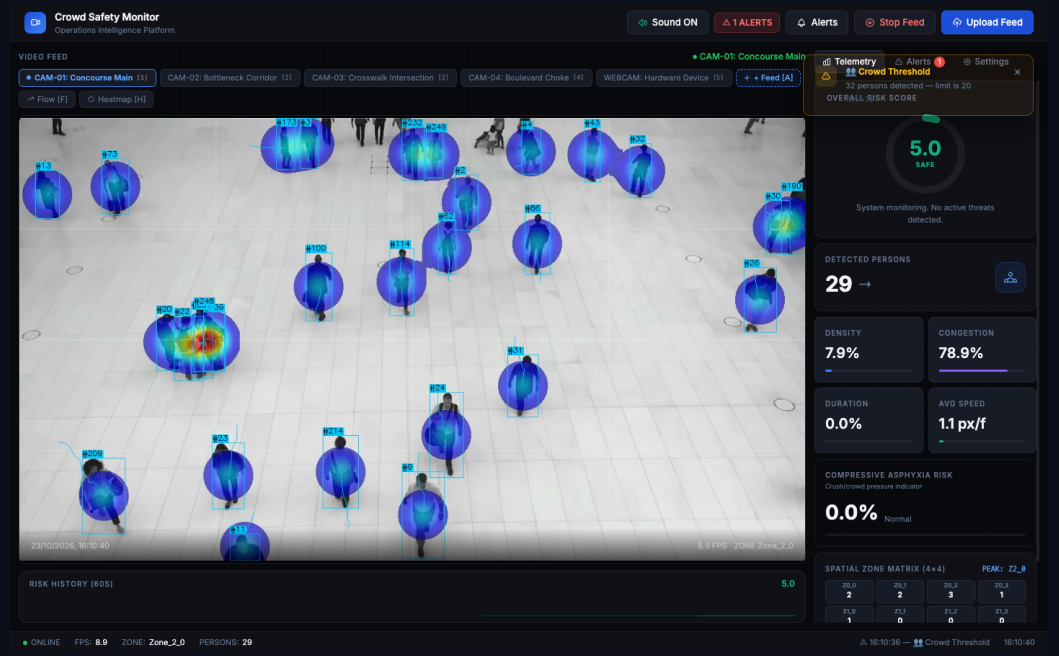
✔ Sub-Pixel Motion Estimation
Vector tails illustrate exact velocity magnitude and direction vectors in real time.

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Spatial Intelligence: 4x4 Zone Matrix & Bottleneck Isolation

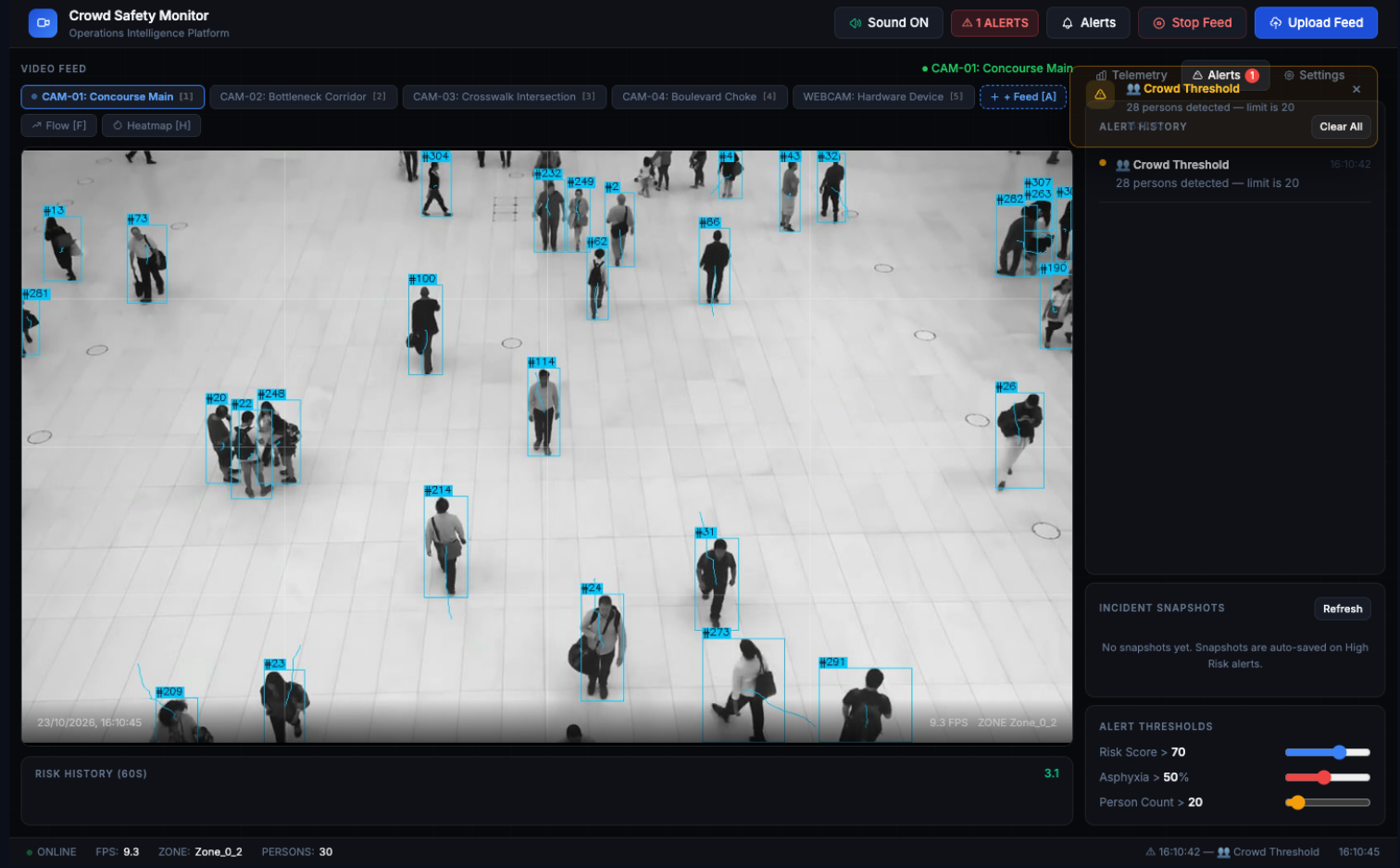
WHY GLOBAL HEADCOUNTS FAIL

- **Localized Choke Points**
  Stampedes happen at bottlenecks (doors, stairs, turnstiles). A global count hides localized life-threatening densities.
- **16 Discrete Micro-Sectors**
  The scene is partitioned into a 4x4 matrix (Z0_0 to Z3_3), monitoring micro-density and speed independently.
- **Peak Zone Isolation**
  System automatically flags the highest-risk zone (e.g. PEAK: Z0_1) for direct security dispatch.
- **Granular Telemetry Stream**
  Every 100ms, all 16 zones broadcast local headcounts, speeds, and densities over WebSockets.



Live 4x4 zone overlay segmenting physical space into discrete risk zones.

Incident Forensics: Automated Snapshots & Alert Logs



CRITICAL RESPONSE AUDITING

Auto-Captured Evidence

When composite risk exceeds threshold (70.0), system automatically freezes and archives high-res annotated JPEGs to disk.

Multi-Tier Risk Badging

Visual notifications display exact zone coordinates, timestamp, and severity level (Warning vs High Risk).

Dual-Tone Acoustic Alert

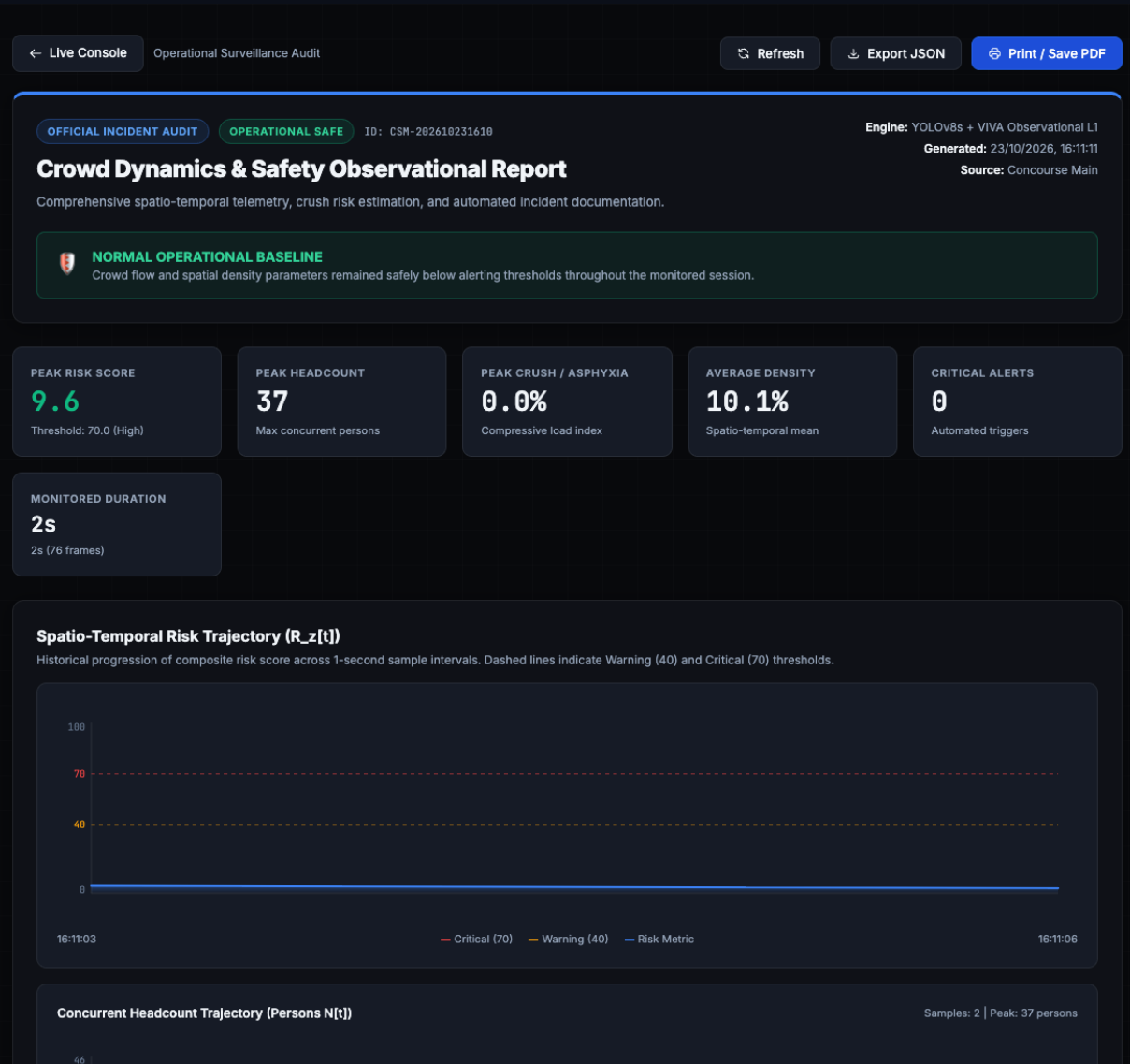
Web Audio API triggers urgent harmonic alarm frequencies to grab operator attention instantly.

Immutable Audit Log

All incidents are timestamped and indexed in disk manifests for official post-event investigations.

CROWD SAFETY MONITOR • OPERATIONS PLATFORM

Post-Incident Dossier: Official Executive Safety Report



AUDIT DOSSIER CAPABILITIES

✔ 1-Click PDF Generation
Generates comprehensive, court-admissible PDF safety audits via headless Chrome printing.

✔ Risk Trajectory Curves
Plots time-series curves of composite hazard score R_z[t] against warning (40) and critical (70) thresholds.

✔ Concurrent Headcount Trajectory
Visualizes influx surges, peak person volumes, and evacuation curve decay.

✔ Session Summary Telemetry
Documents Peak Risk, Peak Headcount, Average Density, and Monitored Frame counts.

Engineering Rigor: Apple Silicon MPS GPU Benchmarks

YOLOv8s INFERENCE

4.5x FASTER

Before: 113.5 ms ➔ Optimized: 25.0 ms

CPU vs Apple MPS GPU

FARNEBÄCK OPTICAL FLOW

3.8x FASTER

Before: 102.9 ms ➔ Optimized: 27.3 ms

1080p Full vs 0.5x Downsampled

END-TO-END THROUGHPUT

4.2x BOOST

Before: 4.8 FPS ➔ Optimized: 20.0 FPS

Unoptimized vs Edge GPU

ADDITIONAL ENGINEERING OPTIMIZATIONS:

- Zero Memory Leaks: Replaced unbounded tracking history lists with collections.deque(maxlen=2000) ring buffers.
- Frame Dropping Prevention: VideoStream runs in decoupled background thread with OpenCV non-blocking reads.

Experimental Validation: Real-World Test Scenarios

SCENARIO 1: HIGH-DENSITY CONCOURSE

CAM-01: Commercial Mall Concourse

Headcount: 38 Persons
Speed: 1.4 px/frame (Free walking)
Spatial Density: 10.0%
Resulting Risk: 9.6 (SAFE)
Key Outcome: Zero false alarms despite dense crowds.

SCENARIO 2: BOTTLENECK QUEUE CHOKING

CAM-02: Temple Entrance Queue

Headcount: 18 Persons
Speed: 0.2 px/frame (Kinetic Lock)
Spatial Density: 48.5% (High localized)
Resulting Risk: 64.2 (WARNING)
Key Outcome: Correctly flags impending crush.

SCENARIO 3: PEDESTRIAN CROSSWALK FLOW

CAM-03: Multi-Lane Crosswalk Intersection

Headcount: 21 Persons
Speed: 1.2 px/frame (Free walking)
Spatial Density: 6.8%
Resulting Risk: 8.0 (SAFE)
Key Outcome: Correctly detects all street-level crossing pedestrians without false alarms.

Competitive Analysis: Comparison with Existing Solutions

FEATURE CAPABILITY	TRADITIONAL CCTV	BASIC AI DETECTION	OUR SYSTEM (CSM)
Real-Time Frame Rate	25-30 FPS (Passive)	5-10 FPS (CPU bound)	**20+ FPS (MPS GPU Edge)**
Spatial Granularity	None (Single screen)	Global bounding boxes	**4x4 Matrix (16 Discrete Zones)**
Immobilization / Velocity Lock	Not measured	Flagged as 'stationary'	**Velocity Decay Curve (Physical Lock)**
Compressive Asphyxia Model	Unsupported	Unsupported	**Non-linear time-pressure integration**
Forensics & Reporting	Manual review	Alert timestamp only	**Automated JPEGs + 1-Click PDF Dossier**

Conclusion & Project Defense Summary

CORE DELIVERABLES & ACHIEVEMENTS

✔ 100% Production-Ready: Completed end-to-end integration across Computer Vision, Telemetry, and Interactive Web HUD.

✔ Zero API Dependencies: Entirely self-contained pipeline runs offline on edge hardware without external cloud latency.

✔ Acoustic & Forensic Evidence: Dual-tone synthesized audio alerts coupled with automatic high-res snapshot capture.

✔ Academic & Mathematical Rigor: Physics-calibrated velocity decay curves and time-integrated asphyxia hazard modeling.

FUTURE ROADMAP & DEMONSTRATION

- **Multi-Camera Re-Identification (ReID)**
  Tracking pedestrian flow across adjacent cameras and corridors.
- **3D LiDAR Point Cloud Fusion**
  Volumetric density estimation invariant to extreme lighting and smoke.
- **Drone Aerial Surveillance Integration**
  Rapid-deployment edge monitoring for open-air religious and sporting festivals.
- **Live Demonstration Ready**
  System is actively streaming live at localhost:8080 with multi-camera controls.